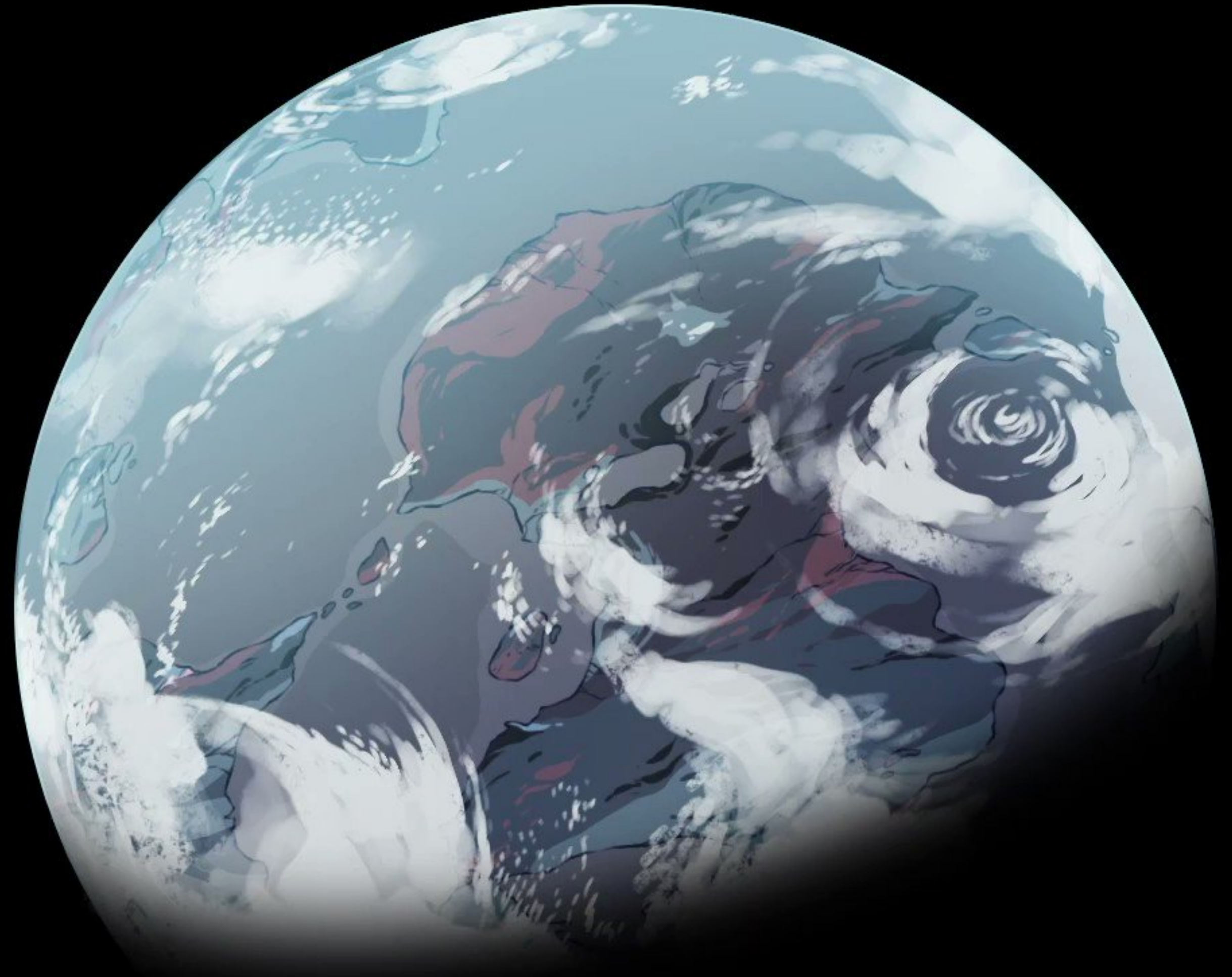


Science & Science Fiction

33-120 | Prof. Katie Breivik



A Trip to the Moon

Georges Méliès (1902)



Le Voyage Dans La Lune: a critique

Realistic

- Bullets and guns are real
- Seeing the sky really bright makes sense without light pollution
- Rocket shape
- Moon is very cold
- Craters
- Casting barrel of gun
- Moon people are light because of less gravity

Possible/Prediction

- Rocket landing in the water
- With enough power the angle of the barrel was possible
- Retrieving floating rocket

Impossible

- Bullets will not make it to the moon
- No astronaut suits
- Umbrellas $\neq$ mushrooms
- Rocket would not just fall back to earth
- Women didn't go to the moon first/moon people in general
- Forests on the moon

Reference Frames: calculating accelerations

As long as an object's velocity stay's constant, it *feels* no acceleration ==> it is in an ***inertial reference frame***

acceleration is a physical quantity that determines how an object's velocity changes with time

$$a = \frac{\Delta v}{\Delta t} = \frac{|v_{\text{start}} - v_{\text{end}}|}{|t_{\text{start}} - t_{\text{end}}|}$$

Objects subjected to acceleration are in ***non inertial reference frames***

Non Inertial Reference Frames

when considering non inertial reference frames

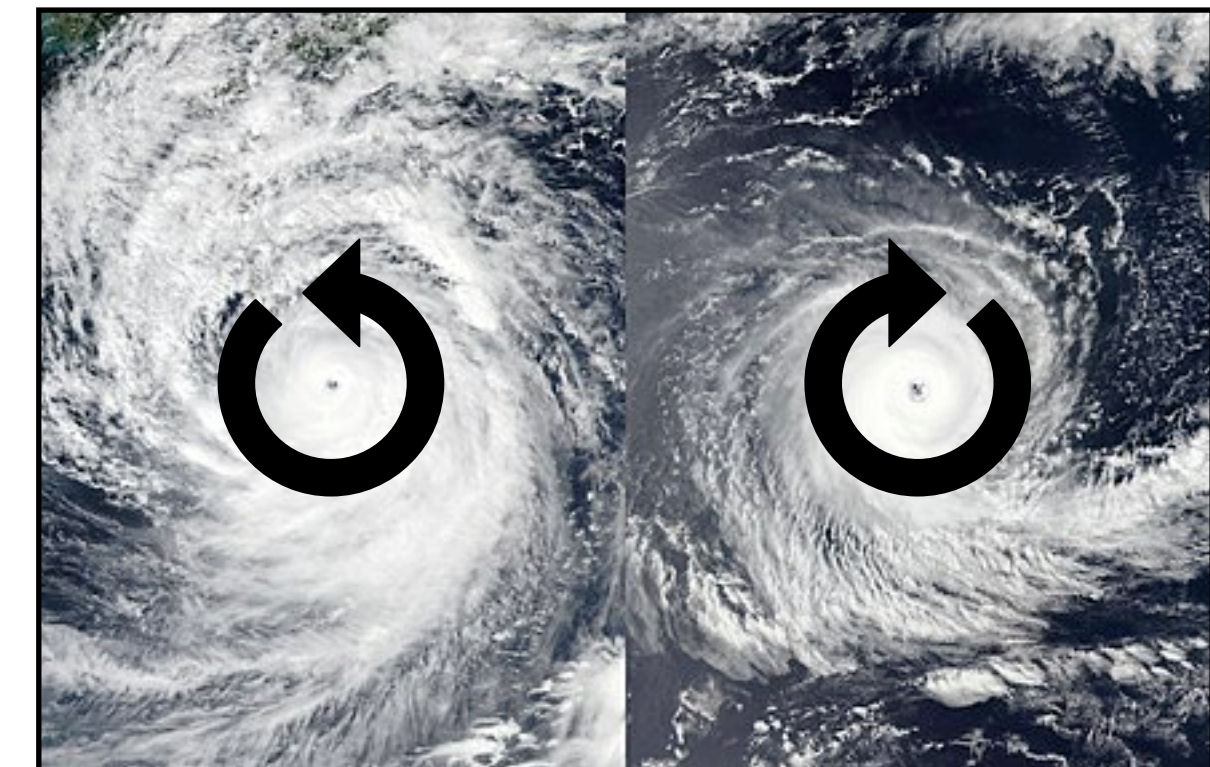


$$a_{\text{gravity}} = 9.8 \text{ m/s}^2$$

scale
matters

$$a_{\text{gravity}} \gg a_{\text{coriolis}}$$

Northern Southern
Anti-clockwise clockwise



$$a_{\text{coriolis}} = 0.03 \text{ m/s}^2$$

Reference Frames: putting a scale to acceleration

human bodies expect to be under 1g on Earth: $1g = 9.8 \text{ m/s}^2$

a standard NASA rocket launch: 3g's

normal human loss of consciousness: 10g's

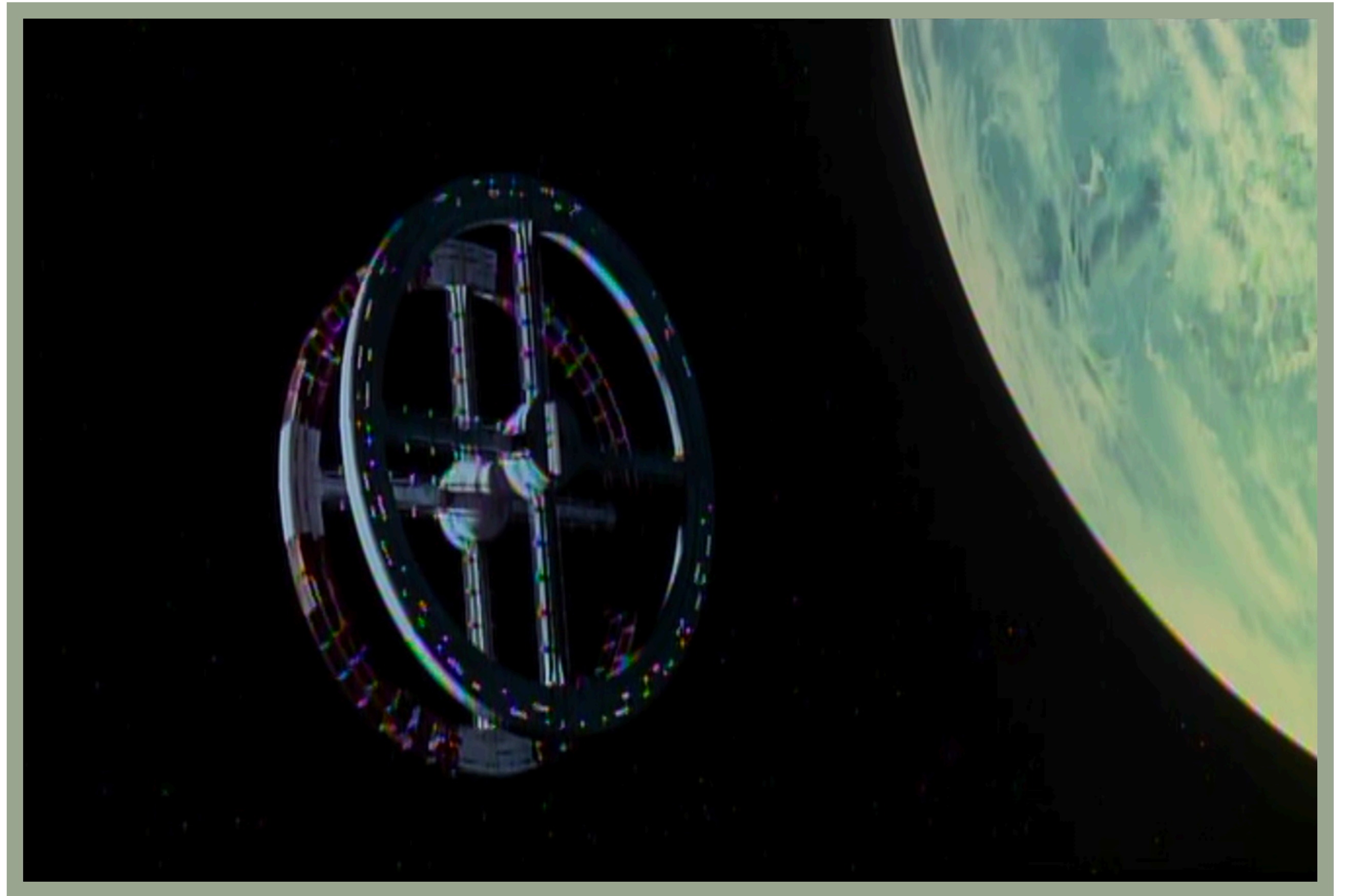
usually lethal: ~20g's

largest g force survived by a human*: 214g's

* Kenny Bräck's 2003 Indycar crash @ 220 mph; F1 crashes are routinely >20g

2001: A Space Odyssey

directed: Stanley Kubrick
Stanley Kubrick
Productions (1968)

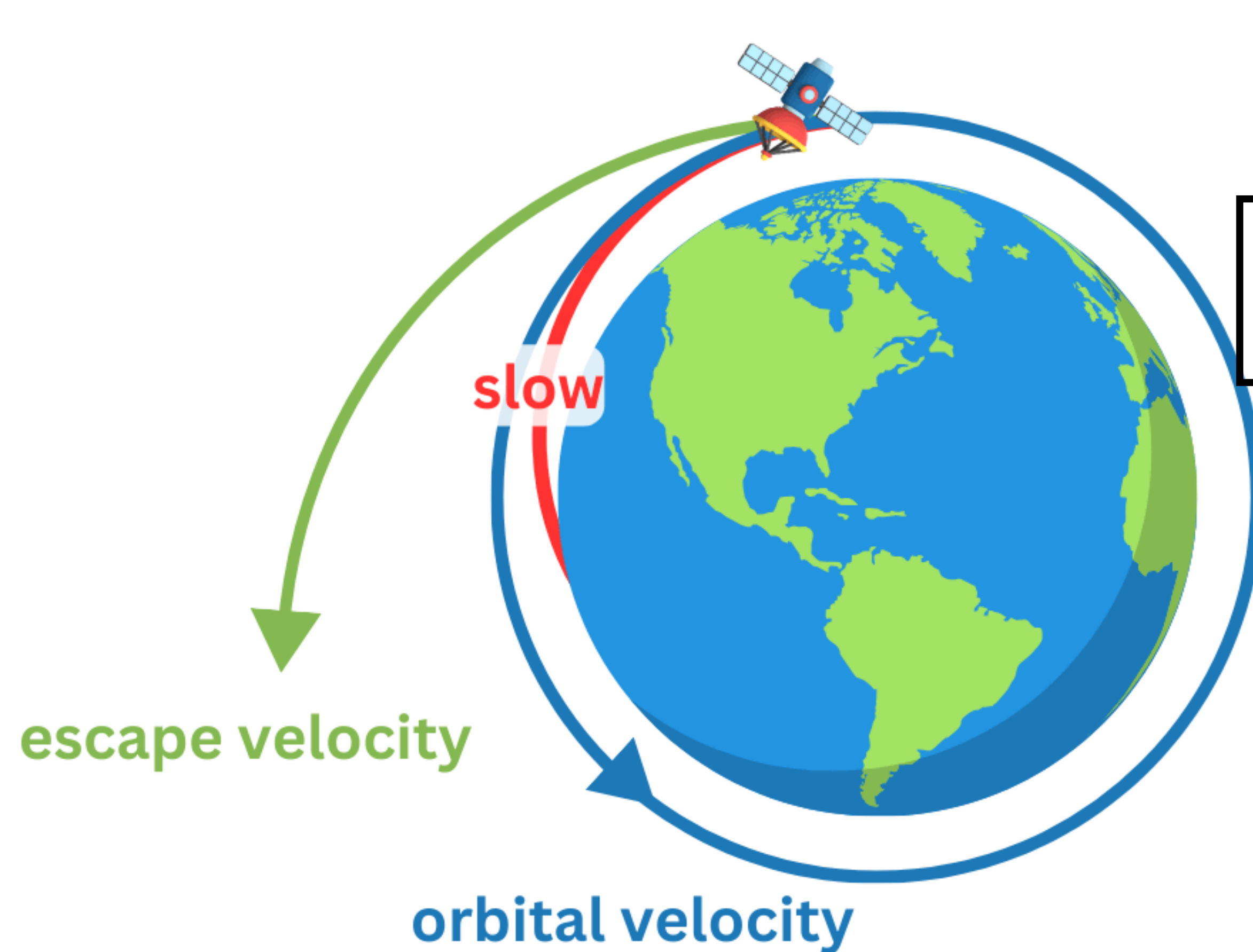


The Martian

directed: Ridley Scott
20th Century Fox (2015)



Escape speed / velocity



gravitational potential energy: mgR

g : acceleration
due to gravity

kinetic energy: $\frac{1}{2}mv^2$

$$\frac{1}{2}mv^2 = mgR$$

$$v^2 = 2gR$$

$$v_{\text{esc}} = \sqrt{2gR}$$

credit: sciencenotes.org

Objects with different masses have different acceleration's due to gravity

$$v_{\text{esc}} = \sqrt{2gR} \quad \leftarrow \text{ This changes their escape velocity}$$

$$g = \frac{GM}{R^2}$$

$$G = 6.67 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$$

$$M_{\text{Earth}} = M_{\oplus} = 5.97 \times 10^{24} \text{kg}$$

$$R_{\text{Earth}} = R_{\oplus} = 6.38 \times 10^6 \text{m}$$

$$g_{\text{Earth}} = \frac{(6.67 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2})(5.97 \times 10^{24} \text{kg})}{(6.38 \times 10^6 \text{m})^2} = 9.78 \text{m/s}^2$$

Objects with different masses have different acceleration's due to gravity

$$v_{\text{esc}} = \sqrt{2gR} \quad \leftarrow \text{ This changes their escape velocity}$$

$$g = \frac{GM}{R^2} \quad g_{\text{Earth}} = 9.78 \text{m/s}^2$$

$$\begin{aligned} v_{\text{esc,Earth}} &= \sqrt{2g_{\text{Earth}}R_{\text{Earth}}} = \sqrt{2 \times 9.78 \text{m/s}^2 \times 6.23 \times 10^6 \text{m}} \\ &= 11,171 \text{ m/s} \simeq 11.2 \text{km/s} \end{aligned}$$

So what acceleration does an astronaut feel when they are launched off Earth?

$$a = \frac{\Delta v}{\Delta t} = \frac{|v_{\text{start}} - v_{\text{end}}|}{|t_{\text{start}} - t_{\text{end}}|}$$

$$v_{\text{start}} = 0 \text{ km/s}$$

$$v_{\text{end}} = v_{\text{esc,Earth}} = 11.2 \text{ km/s}$$

How long for the rocket to reach escape velocity?

$$\Delta t \simeq 6.5 \text{ min} = 390 \text{ s}$$

$$a = \frac{\Delta v}{\Delta t} = \frac{11200 \text{ m/s}}{390 \text{ s}} = 28.72 \text{ m/s}^2 = 2.9 g_{\text{Earth}}$$

So what acceleration does an astronaut feel when they are launched off Earth?

$$a = \frac{\Delta v}{\Delta t} = \frac{|v_{\text{start}} - v_{\text{end}}|}{\Delta t} \quad v_{\text{start}} = 0 \text{ km/s} \quad = 11.2 \text{ km/s}$$

Problem set 1 deals with these kinds of calculations for different contexts

$\Delta t \simeq 6.5 \text{ min} = 390 \text{ s}$

$$a = \frac{\Delta v}{\Delta t} = \frac{11200 \text{ m/s}}{390 \text{ s}} = 28.72 \text{ m/s}^2 = 2.9 g_{\text{Earth}}$$

Next time:

- ▶ Begin chapter 2
- ▶ Newton's laws

Reminder:

- Quiz 1: available on Canvas starting on Friday
- Sci-Fi Problem 1: acceleration to/from Mars
 - available on Canvas starting on Friday
 - due Friday (01/24)